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Review

Polo-Like Kinases

A Team in Control of the Division

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ABSTRACT

Polo, the founding member of the family of polo-like kinases (Plks) was identified in a *Drosphila* screen for mutants affecting spindle pole behavior.[1] Several mutants showed defects at their spindle poles and were hence named after the magnetic poles of the earth or geo-magnetic phenomena associated with them, like Polo and Aurora. Currently, the conserved family of Plks consists of many members throughout various species. Multiple Plks are present in mammalian cells (Plk1, Plk2/Snk, Plk3/Fnk/Prk, and Plk4/Sak) and *Xenopus* (Plx1-3), whereas in other species only one member has been identified, like Polo in *Drosophila*, Cdc5 in budding yeast and Plo1 in fission yeast. Plks are now viewed as important regulators of multiple functions before and during the mitotic cell division. In this review, we will focus our attention on human Plk1 and its family members Plk2-4 and the many roles they play during mitosis. Furthermore, we will describe the currently knowledge of the regulation of these functions.

THE MULTIPLE FUNCTIONS OF Plk1

Expression, activity and localization of Plk1 are all dynamically regulated during the cell cycle. Plk1 expression increases from late S phase to mitosis, a time when Plk1 is most active.[2-4] Plk1 expression is mostly found in highly proliferating tissues.[5] Overexpression of murine Plk1 results in oncogenic transformation, indicating a prominent role for Plk1 in cell division.[6] Degradation of Plk1 by the APC/C starts in late mitosis and continues throughout G_1.[7] Plk1 localization is very dynamic. Plk1 localizes to the cytoplasm and the nucleus during G_2 and is specifically targeted to the centrosomes.[2,8] During early mitosis, Plk1 can be found at centrosomes and kinetochores[2,9] and Plk1 persists there until its degradation. In addition, a fraction of Plk1 translocates to the midzone/midbody late in mitosis.[2,9] Thus, expression, activity and localization of Plk1 are all dynamically regulated before and during mitosis. The multiple sites of Plk1 localization reflect the many functions that Plk1 performs during the cell cycle (Fig. 1).

Mitotic entry. Promoting mitotic entry is one of the numerous functions attributed to Plk1. Initiation of mitosis requires nuclear translocation and activation of Cyclin B/Cdk1. Cyclin B/Cdk1 activation results from the simultaneous inhibition of Wee1 and Myt1 kinases and the activation of Cdc25 phosphatases (reviewed in ref. 10). Plk1 promotes mitotic entry by activating Cyclin B/Cdk1 at all these different levels. (1) Plk1 activates at least one member of the Cdc25 family of phosphatases, Cdc25C. Cdc25C is phosphorylated by Plk1 in a nuclear export signal sequence, thereby leading to nuclear translocation and activation of Cdc25C.[11,12] (2) At the same time, Plk1 phosphorylates Cyclin B/Cdk1-inhibiting kinases Wee1 and Myt1,[13,14] and at least for Wee1, leading to enhanced association with the SCF/beta-TrCP E3 ubiquitin ligase, thereby inducing its degradation.[13] (3) Furthermore, Plk1 phosphorylates Cyclin B at centrosomes, the first site at which Cyclin B/Cdk1 is activated.[15,16] However, whether phosphorylation triggers nuclear import of Cyclin B or activates Cyclin B/Cdk1 complexes in a different way is a matter of debate.[15,16] At least in *Xenopus*, Plk1 and Cyclin B/Cdk1 may act in a positive feedback loop ensuring complete activation of Cyclin B/Cdk1 and mitotic entry.[17] Cdk1 can also phosphorylate Plk1 in an in vitro kinase assay, but this only marginally increases Plk1 kinase activity.[3] The functional relevance of Cdk1-mediated phosphorylation is currently unknown.

The G_2 DNA damage checkpoint. Since Plk1 is important for activation of Cyclin B/Cdk1, one can imagine a possible involvement of Plk1 in restarting the cell cycle after a G_2 arrest by the DNA damage checkpoint. Indeed, the activity of Plk1 is inhibited after

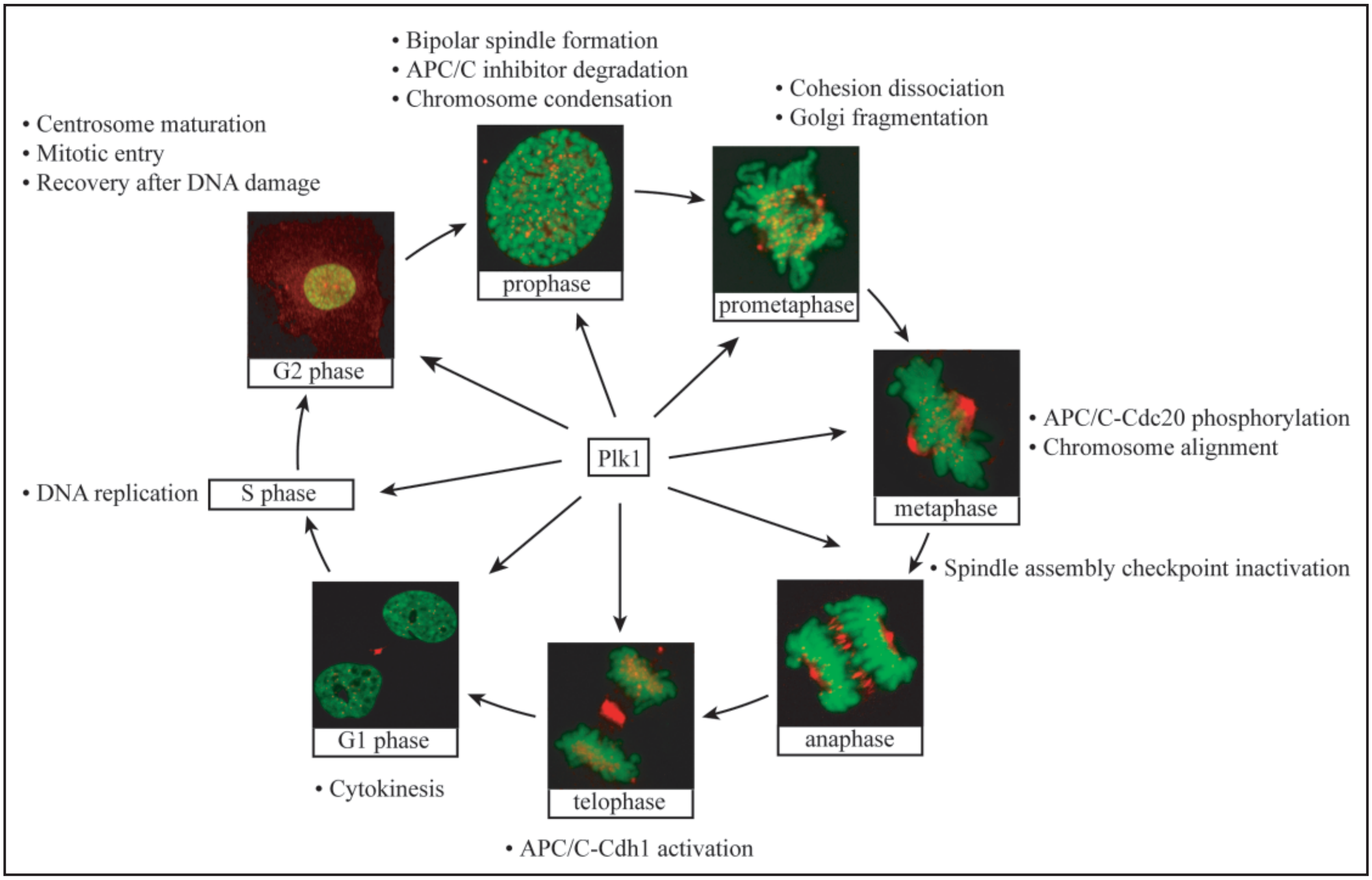

Figure 1 Schematic overview of the different functions of Plk1 during the cell cycle. Indicated are the implications functions of Plk1 at the different stages of the cell cycle and the mitotic phases.

DNA damage and constitutively active mutants of Plk1 can overcome a DNA damage-induced arrest in G_2.[18] Experiments using Plk1 siRNA showed that Plk1-mediated Wee1 degradation is crucial for mitotic entry following recovery from DNA damage.[19] Other targets of Plk1 that are involved in the DNA damage checkpoint have been described, for instance p53.[20] Plk1 can bind to p53 and inhibit its transactivation activity as well as its pro-apoptotic function.[21] Plk1 also interacts with and phosphorylates Chk2.[22] After DNA damage in G_2 and M phases, the association between Plk1 and Chk2 is abrogated.[23] Another substrate of Plk1 involved in the DNA damage checkpoint is the breast cancer susceptibility protein, BRCA2, which is essential for the repair of DNA double-strand breaks.[24,25] Thus, in response to DNA damage, Plk1 activity is inhibited, leading to activation of p53 and modulation of Chk2 and BRCA2, while in recovery from a DNA damage-induced arrest Plk1 may inhibit all 3 proteins.

Centrosome maturation and bipolar spindle formation. Plk1 is also involved in centrosome maturation and separation. The founding member of the Plk family of kinases, Polo, was identified as a *Drosophila* mutant with defective, immature spindle poles.[1] The maturation of centrosomes was also found to depend on Plk1 in human cells, as shown by decreased recruitment of various proteins like γ-tubulin.[26] Plk1 phosphorylates Nlp, a centrosome protein involved in microtubule nucleation. An Nlp mutant lacking Plk1 phosphorylation sites severely disturbs mitotic spindle formation.[27] Nlp is targeted to the centrosome by dynactin and is displaced from this organelle upon mitotic entry.[27,28] Phosphorylation of Nlp by Plk1 inhibits its association with dynein, leading to removal of Nlp from the centrosome,[27-29] probably allowing the recruitment of a mitotic microtubule nucleation scaffold. Concomitantly, Plk1 antibody injection studies show that Plk1 is also essential for centrosome separation, with defects leading to monopolar spindles and consequent mitotic arrest.[26] Plk1 depletion by siRNA can also lead to monopolar spindles.[30,31] Plk1 stably associates with and phosphorylates α-, β- and γ-tubulins and the tubulin-stabilizing protein TCTP,[32,33] suggesting that Plk1 may regulate spindle formation through centrosome maturation and separation as well as by directly regulating tubulin functions. Recently, centrosome recruitment of Plk1 was shown to be dispensable for centrosome maturation and bipolar spindle formation, suggesting that Plk1 kinase activity does not have to be specifically bound to centrosomes in order to execute its centrosomal functions.[34]

Cohesin dissociation. Plk1 has also been reported to regulate chromosome behaviour. Cohesin is a protein complex that holds sister chromatids together before their separation in anaphase. In prophase, the bulk of cohesion is removed from chromosome arms to facilitate sister chromatid resolution, while the remaining cohesin on arms and centromeres is cleaved by separase at the metaphase-anaphase transition (reviewed in ref. 35). Plk1 can completely remove cohesin from chromosome arms in early mitosis.[36,37] Cohesin phosphorylation by Plk1 was shown to reduce its affinity for chromatin.[36] Phosphorylation of the cohesion subunit SA2 is essential for cohesion dissociation from chromosome arms in early mitosis, whereas phosphorylation of cohesin subunit Scc1 by Plk1 enhances its cleavage by separase at the metaphase-anaphase transition.[38,39] In Plk1-depleted cells, chromosomes can still separate by separase-mediated cleavage when cells are forced to override the spindle assembly checkpoint.[37] Thus, although Plk1 may promote sister chromatid resolution by cohesion removal, Plk1 does not appear to be essential for chromosome segregation. Interestingly, in Plk1-depleted cells chromosomes appear hypercondensed,[19,37] indicating Plk1 may be involved in chromosome condensation. Whether this is due to cohesin defects is currently unclear.

The specific persistence of cohesin at centromeres in early mitosis depends on a protein named shugoshin.[40,41] Phosphorylation of the *Drosophila* shugoshin, MEI-S332, by Polo kinase inhibits its activity and induces its dissociation from centromeres.[42] MEI-S332 indeed remains on centromeres in a Polo mutant, but this does not inhibit sister chromatid separation, indicating the existence of a Polo-independent mechanism of MEI-S332 inhibition.[42] Thus, Plk1 may also induce separase-mediated cleavage of cohesin at centromeres by removing its protector shugoshin.

Chromosome alignment and the spindle assembly checkpoint. Recently, Plk1 was also implicated in regulating chromosome alignment. Chromosome congression defects were observed when Plk1 activity was not properly localized in the cell.[34] Plk1 was also suggested to be required for the stabilization of microtubule-kinetochore interactions by creating tension.[30,31,43] Recent studies may provide a molecular mechanism for this process. Plk1 activity is necessary to generate a signal at kinetochores correlating with the lack of tension, i.e., the 3F3/2 phosphoepitope.[43] Furthermore, Plk1 contributes to the recruitment of key kinetochores proteins, like Hec1/Ndc80,[43] which mediate the attachment of microtubules to kinetochores.[44] Thus, Plk1 probably has several roles at kinetochores to coordinate microtubule tension and attachment. Concomitantly, Plk1 siRNA was also found to partially reduce the association of several kinetochore proteins that are important for checkpoint signaling, like Mad2, Cenp-E and Cdc20.[43] However, the spindle checkpoint is still active in Plk1-depleted cells, probably because these proteins can still be found to localize at kinetochores in these cells.[31,43,45] Interestingly, Plk1 was recently found to be enriched on kinetochores that are not under tension and in *Xenopus*, Plx1 binding to kinetochores was found to depend on activity of the checkpoint kinase xMps1.[43,46] Plk1 may thus be involved in tension signaling and upon proper attachment and tension Plk1 may be partially displaced from kinetochores. In turn, this may lead to stabilization of microtubule-kinetochore attachments and silencing of the spindle assembly checkpoint due to the codissociation of checkpoint proteins.

APC/C activation. Interestingly, Plk1 may also be indirectly involved in the activation of separase, thereby again contributing to chromosome segregation. Separase is activated when its inhibitor securin is targeted for degradation by the anaphase-promoting complex/cyclosome (APC/C) at the metaphase-anaphase transition. The activation of the APC/C at the metaphase-anaphase transition is tightly linked to the spindle assembly checkpoint, which only allows APC/C activation once all chromosomes are properly attached to the spindle. Plk1 may be involved in direct and indirect activation of the APC/C. Direct activation of the APC/C may take place by phosphorylation of several APC/C subunits, together with Cyclin B/Cdk1.[47] Phosphorylation of APC/C subunits may be another nonessential function of Plk1, since APC/C substrate degradation can still take place in Plk1-depleted cells and phosphorylation of APC/C subunit Cdc27 is only partially diminished.[30,31,48] In prophase, Plk1 is involved in the indirect activation of the APC/C by inducing the destruction of APC/C inhibitor Emi1.[49,50] An Emi1 homologue, XErp1/Emi2, was recently identified in *Xenopus* and was found to be regulated in a similar manner by Plx1.[51,52]

Cytokinesis. Several Plk family members including Plk1 have been implicated in cytokinesis, consistent with Plk localization to the midzone/midbody late in mitosis. The first evidence of the involvement of Plk1 in cytokinesis was the identification of the kinesin-like protein CHO1/MKLP-1 as an in vitro substrate of Plk1, which induces microtubule bundling and antiparallel movement in vitro.[4] The *Drosophila* Polo was found to interact with CHO1 homologue Pavarotti, which is implicated in cytokinesis.[53] However, no cytokinesis defects were observed after injection of anti-Plk1 antibodies into HeLa cells.[26] On the other hand, overexpression of Plk1 does give rise to multinucleated cells,[54] indicating the precise level of Plk1 may be important for the correct execution of cytokinesis. Evidence for a role for Plk1 in cytokinesis was also found in overexpression studies with a dominant negative mutant of Plk1, which causes a failure in completion of cytokinesis after forced inactivation of the spindle assembly checkpoint.[55] In Plk1-depleted cells, the early stages of cytokinesis can also clearly take place following spindle checkpoint inactivation by depletion of essential checkpoint components.[30] These data indicate that Plk1 may not be essential for the initiation of cytokinesis and possibly also not the later stages of cytokinesis, although this needs to be examined more closely.

Plk1 has been found to phosphorylate several other proteins that are involved in cytokinesis, like NudC, a component of the dynactin complex. NudC depletion yields multinucleated cells and cells that were still connected through their midbody due to abscission defects.[56] NudC with mutations in its Plk1 phosphorylation sites cannot rescue the cytokinesis defects, suggesting that Plk1 phosphorylation of NudC is essential for the execution of cytokinesis.[56] Plk1-mediated phosphorylation of MKlp2, a kinesin localized to the midbody, is also involved in cytokinesis.[57] Injection of an antibody that prevents phosphorylation of MKlp2 by Plk1 gives rise to binucleate cells, indicating that phosphorylation of MKlp2 by Plk1 is necessary for cytokinesis.[57] Finally, the Rho exchange factor ECT2 is another Plk1 substrate recently implicated in cytokinesis.[58]

Another function of Plk1 during mitosis lies in the fragmentation of Golgi stacks during mitosis by phosphorylating the Golgi-associated protein GRASP65.[59,60] Interestingly, several proteins that dissociate from the Golgi during mitosis are implicated in regulation of cytokinesis, thereby coordinating cytokinetic events to Golgi rearrangement. Nir2 is implicated in the regulation of membrane transport and remodeling of cytoskeletal elements and mainly localizes to the Golgi apparatus in interphase cells, but translocates to the cleavage furrow and midbody during cytokinesis.[61] Nir2 may be important for cytokinesis, since overexpression of a deletion mutant causes the formation of multinucleate cells. Plk1 and Cdk1 can both phosphorylate Nir2.[61] Interestingly, overexpression of a Nir2 mutant which fails to interact with Plk1 affects the completion of cytokinesis,[61] suggesting a role for Plk1 in regulating cytokinesis through Golgi-associated proteins.

In budding and fission yeast, mitotic exit and cytokinesis are controlled by the mitotic exit network (MEN) and the septation initiation network (SIN), respectively (reviewed in ref. 62). The MEN and SIN are networks consisting of various proteins including Plk. Interestingly, the MEN and SIN control cytokinesis, even though they are localized to the poles of the mitotic spindle. Analogous to the MEN and SIN, the mammalian centrosome may also regulate cytokinesis. Mammalian cells without centrosomes fail to complete cytokinesis, even though they can form spindles and undergo anaphase.[63,64] During normal mitosis, the mother centriole transiently translocates to the midbody correlating with cleavage furrow ingression and microtubule depolymerization, while movement away from the midbody correlates with cytokinesis.[65] In this way, Plk1 may be translocated to the midbody, since Plk1 localization at the centrosome indeed diminishes in late mitosis (our unpublished observation). The molecular link between the centrosome and cytokinesis in human cells is starting to be unravelled. Depletion of centriolin or Cep55, components of the mother centriole, gives rise to cytokinesis failure.[66,67] Interestingly, Cep55 was recently identified as a substrate of Plk1 as well as Cdk1/Erk.[67] Cells expressing a phosphorylation-deficient mutant of Cep55 also undergo cytokinesis failure, indicating the importance of phosphorylation in this process. Thus, Plk1 may regulate cytokinesis, possibly through modulation of centrosome-, Golgi- and midbody-associated proteins.

Other functions of Plk1. Besides being involved in mitotic entry, the DNA damage checkpoint, centrosome maturation and separation, bipolar spindle formation, APC/C activation, chromosome segregation, Golgi fragmentation and cytokinesis, Plk1 has been implicated in even more processes. For instance, besides being a substrate of the proteasome, Plk1 can also enhance proteolytic activity of proteasomes through phosphorylation.[68] Furthermore, Plk1 enhances stability of the peptidyl-prolyl isomerase Pin1, a protein important for protein folding and transport.[69] Depletion of Pin1 in HeLa cells results in mitotic arrest, whereas overexpression of Pin1 induces a G_2 arrest, suggesting that Pin1 both negatively regulates mitotic entry and is required for proper mitotic progression.[70] Pin1 inhibits mitotic division in *Xenopus* embryos and entry into mitosis in *Xenopus* extracts.[71] Pin1 specifically recognizes phosphorylated Ser/Thr-Pro bonds present in mitotic phosphoproteins such as Cdc25, Myt1, Wee1, and Cdc27, all known Plk1 substrates, and Plk1 itself.[72] Unlike wild-type Pin1, mutants that cannot interact with mitotic phosphoproteins do not give rise to the above listed mitotic entry and progression defects in *Xenopus*.[71] Thus, Plk1 promotes stability of Pin1, which in its turn acts as an essential regulator of mitotic phospho-proteins.

In budding yeast and mouse cells, a role for Cdc5/Plk has been proposed in DNA replication at origin complexes.[73,74] In contrast, injection of anti-Plk1 antibodies had no effect on the kinetics of DNA replication in HeLa cells, suggesting that Plk1 has no major role in DNA replication in human cells.[26] However, overexpression of a constitutive active mutant of Plk1, Plk1-S137D, during a G_1/S arrest and consequent release into the cell cycle induces S-phase progression defects, suggesting that under certain circumstances Plk1 can influence DNA replication.[75,76] A putative target for Plk1 in this process may be the APC/C, since the assembly of prereplicative complexes at origins only occurs when APC/C activity is high (and Cdk activity is low), while origin firing can only take place when the APC/C is inactivated (and Cdks become active)(reviewed in ref. 77). Thus, overexpression of certain APC/C-activating mutants of Plk1 may stall DNA replication. Whether Plk1 also directly functions in DNA replication has to be further investigated. In budding yeast and *Xenopus*, Cdc5 and Plx1 are involved in adaptation from an S phase arrest after incomplete DNA replication,[78,79] but whether this role is conserved in human cells remains to be determined.

How can one protein be important for this vast list of actions in a time-span of only about one hour during mitosis? Certainly, aspects of Plk1 function will be regulated through its dynamic subcellular localization. Also, several regulatory elements can be found in Plk1 that contribute to the various functions in several ways. Furthermore, Plk1 is post-translationally modified by upstream kinases. We will discuss the current knowledge of these modes of Plk1 regulation below.

THE FUNCTIONS OF Plk1 HOMOLOGUES Plk2-4

Plk2. Plk2/Snk was originally identified as a member of the Plk family through its extensive homology with Plk1. However, its tissue distribution and functions differ greatly from Plk1. A broad tissue distribution can be found of Plk2 transcripts, whereas Plk1 expression is restricted to proliferating tissues such as placenta, colon and testis.[80] Furthermore, whereas Plk1 was reported to function in G_2/M, Plk2 was identified as an early-growth-response gene, which plays important roles in regulating cell proliferation in G_1.[81] In NIH 3T3 cells, mRNA encoding Plk2 was markedly elevated within 1 h after treatment with mitogen.[81] Furthermore, whereas Plk1 is not detected in the brain, Plk2 is constitutively expressed in post-mitotic neurons.[82] Stimuli that produce synaptic plasticity dramatically increase levels of Plk2 mRNAs. Moreover, induced Plk2 proteins are targeted to the dendrites of activated neurons, suggesting that Plk2 mediates phosphorylation of proteins in this compartment.

Plk2 may primarily function as a regulator of G_1 and early S phase progression, a time when both Plk2 mRNA and protein levels peak.[81,83] Overexpression and depletion of Plk2 lead to an increase and decrease of centrosome numbers, respectively, indicating that Plk2 is required for centriole duplication during S phase.[84] Overexpression of a kinase-deficient mutant of Plk2, but not Plk1, in S-phase arrested cells blocks centriole duplication in Chinese hamster ovary and U2OS cells. Thus, Plk1 and Plk2 both regulate centrosome function, albeit at different times in the cell cycle. Plk2 regulates centriole duplication during S phase, after which Plk1 ensures centrosome maturation in late G_2. Plk2 is a nonessential gene. Plk2$^{-/-}$ embryos are viable[85] and Plk2 depletion in HeLa and U2OS cells confirmed that Plk2 is not required for progression through an unperturbed cell cycle,[86] suggesting Plk2 has a non-essential function in centriole duplication. Whether other Plk family members can compensate for loss of Plk2 remains to be determined. However, Plk2 does play a role in cell cycle progression since Plk2$^{-/-}$ embryos do show slightly retarded growth and skeletal development.[85] Cultured Plk2$^{-/-}$ embryonic fibroblasts grow more slowly than normal cells, having a cell cycle of 25 hr instead of 22 hr, and show delayed entry into S phase.[85] Disruption of other cell cycle genes has also been reported to lead to growth retardation while having a normal cell cycle progression e.g., cyclin D1.[87]

Besides functioning in G_1/S, Plk2 possibly also acts later in the cell cycle. Plk2 silencing in the presence of taxol or nocodazole significantly increases apoptosis,[86] suggesting that Plk2 may prevent mitotic catastrophe following spindle damage. So, even though Plk2 is nonessential in an unperturbed cell cycle, it may have important functions in the case of spindle damage. However, the observed effect may also be caused by a failure to duplicate centrioles, which might lead to sensitization of cells to microtubule inhibitors in mitosis, as suggested by (ref. 84). Therefore, it still needs to be confirmed whether Plk2 actually has functions outside G_1/S. Plk2 may also play a role in the DNA damage checkpoint, since Plk2 mRNA expression is induced by DNA-damaging agents.[86,88] In contrast, Plk1 gene expression levels are decreased in the presence of DNA damage,[21,89] suggesting that Plk2 and Plk1 may functionally oppose each other during a DNA damage response. For all above described functions attributed to Plk2, only one neuron-specific substrate has been identified. Plk2 phosphorylation of SPAR (spine-associated Rap guanosine triphosphatase-activating protein), a regulator of actin dynamics and dendritic spine morphology, leads to its degradation and causes loss of mature dendritic spines.[90] The identification of more (nonneuronal) substrates is necessary to understand the mechanisms by which Plk2 functions.

Plk3. Like Plk2, Plk3/Fnk/Prk was identified as an immediate-early gene with mRNA levels peaking shortly after mitogen addition.[91] In contrast to Plk2 however, Plk3 protein levels remain relatively constant during the cell cycle.[9,92,93] Plk3 and Plk2 transcripts have a broad tissue distribution, whereas Plk1 transcripts are specifically found in proliferating tissues.[80] Like Plk2, Plk3 is constitutively expressed in post-mitotic neurons and is probably involved in synaptic plasticity.[82] Plk3, but not Plk2, was suggested to play a role in cellular adhesion.[80] In general, the biological functions of Plk3 and Plk1 appear to oppose each other. For instance, overexpression of murine Plk1 results in oncogenic transformation,[6] whereas ectopic expression of human Plk3 inhibits cell growth by inducing apoptosis.[94] However, both kinases complement Cdc5 temperature-sensitive mutants of budding yeast,[92,95] indicating there is at least some degree of functional overlap between Plk3 and Plk1.

Plk3 has many reported functions in the cellular response to DNA damage. Again, Plk3 and Plk1 appear to perform opposing functions in this process. As described above, Plk1 expression is downregulated after DNA damage[21,89] and relative kinase activity is inhibited through regulation by ATM/ATR.[45] In contrast, Plk3 activity is rapidly increased in response to DNA damage and cellular stress, with expression levels remaining unchanged.[20,96] Plk3 becomes phosphorylated and activated following DNA damage or upon spindle poison addition, but not normally in mitosis.[9] ATM is involved in upregulation of Plk3 kinase activity specifically after DNA damage, but not spindle damage.[9,20]

In the DNA damage response, Plk1 and Plk3 can phosphorylate the same substrates, albeit on different residues, e.g., the tumor suppressor p53.[20] Plk1 physically binds to p53 and inhibits its transactivation activity as well as its pro-apoptotic function.[21] In contrast to Plk1, Plk3 may promote p53 function. Plk3 phosphorylates Ser-20 of p53,[20] a residue which is also phosphorylated by Chk1 and Chk2, resulting in p53 stabilization by interaction with HDM2 and cell cycle arrest.[97-99] These data suggest that Plk3 functionally links DNA damage to cell cycle arrest and apoptosis via the p53 pathway.

Another substrate that Plk3 and Plk1 have in common is Cdc25C. Plk1 phosphorylates Ser-198 of Cdc25C, leading to its activation and nuclear import.[11,12] Conflicting reports exist about the residues phosphorylated by Plk3 and whether phosphorylation leads to nuclear or cytoplasmic accumulation of Cdc25C. In one report, Plk3 was shown to phosphorylate Cdc25C primarily on Ser-191 and to a lesser extent on Ser-198 in vitro. Phosphorylation of these residues located within a nuclear exclusion motif led to nuclear accumulation of Cdc25C.[100] Another report however, showed that a different residue, Ser-216, is the major Plk3 phosphorylation site of Cdc25C in vitro.[101] Chk1 and Chk2 can also phosphorylate Ser-216, leading to 14-3-3 binding and sequestration in the cytoplasm and consequent inhibition of Cdc25C.[102,103] This suggests that Cdc25C phosphorylation on these residues by Plk3 should also result in cytoplasmic localization and inhibition of Cdc25C. It remains to be determined what residues in Cdc25C are phosphorylated by Plk3 in vivo and whether its phosphorylation by Plk3 leads to nuclear or cytoplasmic accumulation. However, during a DNA damage-induced arrest, Cdc25C is kept inactive and Plk3 activity is upregulated, making it more likely that Cdc25C is kept inactive by Plk3 by sequestering it in the cytoplasm.

In regulating p53 and Cdc25C, Plk3 may act parallel to or downstream from Chk2. Chk2 physically interacts with and stimulates Plk3 activity.[104] The interaction of Chk2 and Plk3 is enhanced upon DNA damage, suggesting Chk2 may mediate direct activation of Plk3 in response to genotoxic stresses.[104] In its turn, Plk3 has been reported to contribute to full activation of Chk2 in vivo together with ATM,[9] suggesting a putative positive feedback loop.

Plk3 can phosphorylate at least 4 substrates that have not been reported as Plk1 targets. Plk3 was reported to phosphorylate Cdc25A and Cdc25B in vitro[101] and like Plk2, Plk3 also phosphorylates neuron-specific SPAR, leading to its degradation and consequent loss of mature dendritic spines.[90] Plk3 also phosphorylates DNA polymerase δ, suggesting that Plk3 may be involved in DNA damage repair.[105] Possible other interaction partners of Plk3 are the C-terminus of Plk1, the catalytic subunit of protein phosphatase 1α and Tensin2, a focal adhesion molecule that binds to actin filaments and positively regulates cell migration.[104] It will be interesting to examine whether these are true substrates or regulators of Plk3, especially to establish a possible direct link between Plk3 and Plk1.

Whether Plk3 is essential for any of the above described functions is unknown. So far, no knock-out mouse or cells depleted of Plk3 have been reported, even though Plk3 siRNA has been used to reverse the nuclear accumulation of Cdc25C after Plk3 overexpression.[100] Examining the phenotype of Plk3-depleted cells or mice will yield valuable data concerning the functions of Plk3 and a possible redundancy between different family members.

Plk4. Plk4/Sak/Stk18 was the last member of the human Plk family to be identified. Murine Plk4 mRNA levels increase late in G_1 phase and remain elevated through S phase and mitosis, before declining early in G_1.[106] Whether Plk4 expression and kinase activity are regulated in a similar manner is presently unknown. However, it was reported that Plk4 expression is associated with mitotic and meiotic cell division in mouse tissues.[107]

Like Plk1, Plk4 is essential for cell viability. Expression of a murine Plk4 antisense fragment in CHO cells suppresses cell growth[107] and Plk4 siRNA in HeLa cells induces apoptosis, like Plk1.[108] Plk4$^{-/-}$ embryos arrest after gastrulation at E7.5 with an increase in mitotic and apoptotic cells.[109] Plk4$^{-/-}$ embryos display mitotic cells with high levels of Cyclin B, suggesting that Plk4 is required for the APC/C-dependent destruction of Cyclin B and for mitotic exit in the postgastrulation embryo.[109] Compared to wild-type cells, heterozygous Plk4$^{+/-}$ embryonic fibroblasts grow slower and have increased centrosomal amplification, multipolar spindle formation and aneuploidy.[110] Overexpression of murine Plk4 has been reported to inhibit colony-formation in CHO cells and to increase the number of multinucleated cells.[106]

Exact functions of Plk4 remain largely unknown. Recently, Plk4 was reported to function in centriole duplication in U2OS and HeLa cells in cooperation with Cdk2.[111,112] It was suggested that this finding provides an attractive explanation for the crucial function of Plk4 in cell proliferation.[111] Plk2 is also required for centriole duplication, but is not essential for proliferation (see above), suggesting that Plk2 function at the centrosome is either not essential or that other proteins like Plk4 can substitute for Plk2. Alternatively, yet unidentified functions of Plk4 may contribute to the essential role of Plk4 in the cell cycle. Identifying other functions or interaction partners of Plk4 will advance our understanding of this interesting Plk family member.

STRUCTURAL FEATURES OF Plks

Plk family members are highly homologous in their catalytic domain in the N-terminus and in C-terminal motifs named the Polo boxes. The kinase domain is responsible for kinase activity of Plk1, whereas the Polo boxes serve as protein-protein interaction motifs that bind substrates and interactors and target Plk1 to specific subcellular structures.

The Plk1 kinase domain. Up to date, the structure of the kinase domain has not been solved. However, several studies using point-mutants have provided us some information about important residues in the kinase domain (Fig. 2). Most of these residues were identified because of sequence homology with regulatory residues of other kinases. For instance, Plk Asp-194 was found to be important for kinase activity through homology with the DFG sequence of kinase subdomain VII in Cdks, which gives a dominant-negative enzyme when mutated in Cdks.[113,114] Mutation of Plk Asp-194 to Arg or Asn indeed abolished kinase activity in budding yeast and murine Plk.[95,115] Other kinase deficient mutants were generated by mutating Plk1 Lys-82 in the putative ATP-binding site,[4] Asn-172 in Plx1,[116] murine Plk Glu-206 and the activation loop residue Thr-210 in *Xenopus* and mammalian Plk.[75,95,117] Overexpression of kinase-deficient Plk leads to a delayed mitotic entry and consequent mitotic arrest due to substrate competition with endogenous Plk1.[55,118-120] Defects in proper bipolar spindle formation were observed, even though centrosome maturation appeared normal.[55]

Although the crystal structure of the kinase domain is unknown, it has been predicted through modeling based on the crystal structures of Aurora A, Aurora B and Akt/PKB.[121] The resulting model shows that Thr-210 is located in the activation loop. Phosphorylation of an analogous Thr in PKA opens and stabilizes the conformation to allow substrate binding,[122,123] suggesting a similar conformational change may take place after Thr-210 phosphorylation. Ser-137 localizes to the predicted substrate-binding cleft and phosphorylation of this residue likely changes the substrate specificity of Plk1.[121] Studies of phospho-mimicking mutants of Ser-137 and Thr-210 have been described. Cells overexpressing these constitutive active Plk1-T210D or S137D prematurely enter mitosis.[76] Expression of Plk1-T210D leads to a mitotic delay, which depends on the spindle assembly checkpoint.[75,76] In contrast, cells inducibly expressing Plk1-S137D prematurely activate the APC/C independent of the spindle checkpoint.[76] However, when Plk1-S137D is overexpressed during an arrest in G_1/S phase, cells arrest or undergo a delay in S-phase upon release from the arrest, suggesting a possible role for Plk1 during DNA replication.[75,76]

Plk1 does not only phosphorylate other proteins, it can also phosphorylate itself.[2,3] In *Xenopus* Plx1, Ser-260 and Ser-326 are residues that are probably auto-phosphorylated, since neither residue is phosphorylated upon incubation of M phase *Xenopus* extracts with a kinase-defective Plx1.[124] These residues are conserved in Plk1 at Ser-269 and Ser-335, but whether they are also auto-phosphorylation sites remains to be determined. Auto-phosphorylation may enhance general Plk1 activity, but the exact function of auto-phosphorylation has not yet been determined.

The Plk1 polo-box domain. The region containing the 2 Polo boxes is also known as the Polo-box domain (PBD). The PBD is a single functional motif that regulates cellular function, subcellular localization and substrate interactions.[55,125,126] Insight into how PBD-protein interactions are regulated came from a proteomic screen for phospho-Serine and phospho-Threonine binding motifs.[125] This screen identified the PBD of Plk1 as a pSer/pThr-binding motif, providing a molecular model in which priming phosphorylation directs Plk1 substrate recognition and activation at the right time to the right place. Mutation of the putative Plk1 pThr-binding motif in Cdc25C indeed abolishes interaction between Cdc25C and the Plk1-PBD, and decreases phosphorylation of Cdc25C.[125]

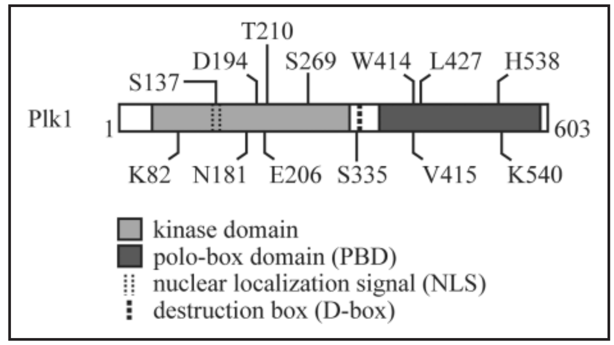

Figure 2. Schematic representation of residues important for the function of Plk1. See text for additional information about the function of these Plk1 residues.

The crystal structure of the PBD showed that its pSer/pThr-binding site is formed by a shallow cleft between the 2 Polo boxes.[127,128] The only residues that contact the phosphate group directly are His-538 and Lys-540. Plk1-H538A/K540M indeed eliminates phosphopeptide binding of the PBD in vitro, Cdc25C binding in vivo and localization to centrosomes,[127] indicating the importance of these residues for Plk1 functions.

Residues Trp-414, Val-415 and Leu-427 all contribute to proper Plk localization and function in various Plk family members, including budding yeast Cdc5, fission yeast Plo1, murine Plk and human Plk1.[55,120,126,129-131] Mutation of the residue analogous to Trp-414 in *Xenopus* Plx1 and Val-415 in budding yeast Cdc5 inhibits interaction with substrates, such as the APC/C subunit Cdc27.[120] Trp-414 indeed fulfills a critical role in ligand binding, as revealed by the crystal structure of the PBD.[127,128] Val-415 and Leu-427 appear not directly involved in ligand interactions, but may maintain structural stability.

Since the PBD is necessary for substrate binding and subcellular localization, one can hypothesize that its overexpression may act in a dominant-negative fashion to inhibit the actions of endogenous Plk1. Indeed, overexpression of the C-terminus of Plk1 causes mitotic arrest and spindle stability defects.[55,118]

Intramolecular interaction in Plk1. Evidence for intramolecular interactions between the kinase domain and the PBD was first found in Plk1 deletion experiments. Plk with a deleted C-terminus possesses several-fold increased kinase activity, suggesting that the PBD can inhibit the kinase domain.[54,95,118] Indeed, the C-terminus of murine Plk can bind to the catalytic domain of Plk and inhibit its kinase activity.[118] The phosphopeptide-binding site in the PBD is not involved in this interaction, since the kinase domain can still bind to Plk1-H538A/K540M.[127] Kinase activity of Plk1 is stimulated by phospho-peptide binding of the PBD, possibly through a conformational change that relieves internal inhibition.[127] However, the PBD was found to behave as a rigid structure and no conformational changes were detected in the PBD upon peptide binding.[128] These

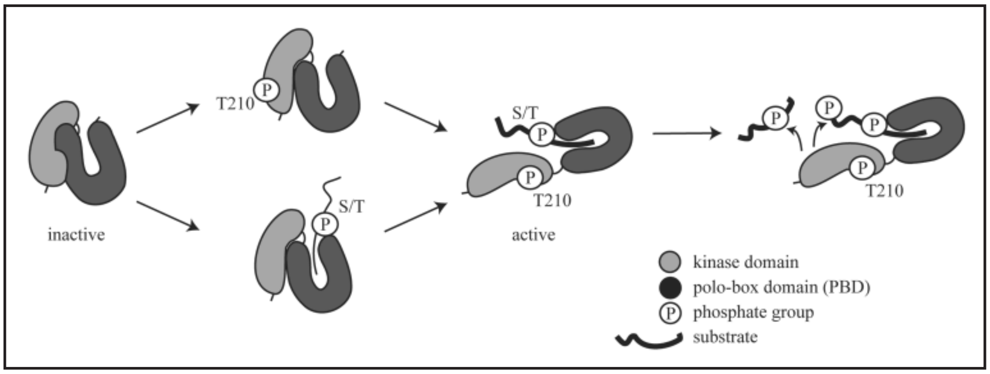

Figure 3. Model for Plk1 activation and substrate phosphorylation. In the active state, the kinase domain and the PBD of Plk1 bind intramolecularly and inhibit each other. Plk1 can be activated to phosphorylate its substrates in two, not mutually exclusive ways. (i) the kinase domain is phosphorylated e.g., on Thr-210, which abolishes inhibition by the PBD and increases kinase activity. (ii) The PBD binds to a phosphorylated substrate, leading to a consequent increase in kinase activity. When internal inhibition is relieved by one or both of these events, Plk1 is activated to target its substrates.

data suggest that substrate binding by the PBD not only recruits the catalytic domain to substrates but increases its activity at the same time, ensuring that full activation of Plk1 towards its substrates is permitted only in the right place at the right time.

In its turn, the kinase domain inhibits phospho-peptide binding by the PBD as well, suggesting that a mutually inhibitory interaction exists.[127] Indeed, binding of the PBD to the kinase domain only occurs with unphosphorylated Plk from G_2 cells, but not with phosphorylated, active Plk from mitotic cells or when Thr-210 in the activating T-loop is mutated into Asp, which elevates kinase activity.[75] These data suggest that activation of the kinase domain in general or phosphorylation of Thr-210 in specific can release the activation loop from C-terminal inhibition. The effect of Thr-210 modification and deletion of the C-terminus are not additive, suggesting that phosphorylation and C-terminal deletions have, in part, a similar influence on catalytic activity.[75] Thus, inhibition of the kinase domain by the PBD can be relieved both by substrate binding of the PBD and phosphorylation of Thr-210 (Fig. 3). Which event serves as the initial Plk1 activation trigger is an intriguing question waiting to be answered.

Other structural features of Plk1. In G_2, Plk1 accumulates in both the cytoplasm and the nucleus.[8] Nuclear targeting is mediated by a conserved region in the kinase domain of Plk1, the bipartite nuclear translocation signal (NLS).[8] Residues 134–136 and 143–146 are both necessary and sufficient for nuclear localization of Plk1 in HeLa cells.[8] Expression of an NLS-disrupted mutant during S phase was found to arrest cells in G_2, suggesting that nuclear localization of Plk1 before mitosis is important for cell cycle progression.[8]

A putative destruction box was identified in Plk1 in between the kinase domain and the PBD.[132] This RxxL-like motif spans from residue Arg-337 until Lys-345. Mutation of the conserved RxxL residues in this motif into Alanine yields a Plk1 mutant that is stable in anaphase and that delays mitotic exit.[132]

Structural motifs in Plk2-4. The kinase domain and the PBD are the regions with highest homology in Plk1 homologues. There is little similarity in the linker region between the kinase domain and the PBD (amino acids 367-410). Structural results show that Plk1-3 are very similar in the core of the PBD, unlike Plk4.[128] In Plk1-3, 9 of the 12 residues most intimately involved in substrate binding are identical, including Trp-414, His-538 and Lys-540, and conservative changes are found in most of the other residues,[128] suggesting similar substrate binding properties. Indeed, Plk2 and Plk3 PBDs were found to preferentially bind pSer/pThr-containing motifs, like Plk1.[127] Plk4 differs from Plk1-3 in that it has only a single Polo box instead of 2. Although the Plk4 Polo box can form intermolecular dimers, Plk1-3 and Plk4 Polo boxes are structurally distinct. The different architectures argue against conservation of phosphoprotein binding function since residues that are most involved in phosphopeptide binding by Plk1 are not conserved in Plk4.[127,128] Therefore, it seems unlikely that the Plk4 PBD can also function as a phospho-protein binding motif. However, the Plk4 PBD probably is instrumental in proper subcellular localization of Plk4. At least for mouse Plk2 and 4, centrosome localization depends on the PBD.[83,109] The process of intramolecular inhibition is conserved in Plk2, since the C-terminus of Plk2 was shown to inhibit kinase activity.[86] Further research is needed to investigate whether this internal inhibitory mechanism is also present in Plk3 and Plk4, and whether Plk3 localization is dependent on the PBD.

If the PBDs from Plk1-3 all bind pSer/pThr-containing motifs, then how can their cellular functions differ? Many different explanations exist: (1) their expression is regulated differently (see above), (2) they may be activated at different times or places by upstream kinases, (3) substrate specificity of their kinase domain may differ, (4) the unconserved linker region between the kinase domain and the PBD may be involved or (5) other, unknown targeting regions may influence their functions.

Most of the important residues in the kinase domains of Plk2-4 are conserved, like Lys-82, Asn-181 (analogous to Plx1 Asn-172), Asp-194 and Thr-210. Glu-206 can only be found in Plk4, whereas Ser-137 is present in Plk2-3, but not Plk4. Neither of the putative auto-phosphorylation sites Ser-269 and Ser-335 is conserved in Plk2-4.

The bipartite NLS is largely conserved in Plk2-3. In contrast, only the second basic cluster of the NLS is partly conserved in Plk4, which is also the case in budding and fission yeast Plk. The D-box, the target site for APC/C-mediated destruction, is not conserved in Plk2-4. Nothing is known about the mechanism of Plk2 degradation, but since Plk2 expression peaks in G_1/S Plk2 is more likely degraded through the SCF. Murine Plk3 was shown to be degraded by the ubiquitin-proteasome system when overexpressed,[133] but whether endogenous Plk3 is similarly regulated remains unknown. Plk4 has a similar expression pattern as Plk1 and is also polyubiquitinated,[106] suggesting the APC/C may also be involved in its degradation. However, Plk4 is a highly unstable protein by itself.[107] It contains several PEST sequences which are critical for this instability, since deletion of these sequences stabilizes expression.[134] Plk4 can be protected from PEST sequence-dependent proteolysis by a protein tyrosine kinase named Tec,[134] indicating Plk4 degradation may be regulated by post-translational modifications instead of the APC/C.

REGULATION BY UPSTREAM FACTORS

Regulation of Plk1. Compared to the long list of reports on Plk1 functions, relatively few studies on upstream regulators of Plk1 exist. We do know that Plk1 actions are regulated by different phosphorylation events,[3,4,54] but it is still relatively unclear what kinases are involved. Several kinases have been reported as upstream regulators of Plks in other species, e.g., Cdk1, MAPK and PKA. At different embryonic phases in starfish, distinct upstream kinases for Plk1 were identified, i.e., Cyclin B/Cdk1 at meiosis I, MAPK as well as Cyclin B and/or Cyclin A/Cdk1 at meiosis II and Cyclin A/Cdk1 at embryonic M phase.[135] The spindle assembly checkpoint kinase xMps1 was also reported to phosphorylate Plx1, thereby regulating the binding of Plx1 to kinetochores.[46] In human cells, one report shows that Cdk1 can phosphorylate Plk1 in an in vitro kinase assay,[3] but in another report Plk1 was found to not be directly regulated by either Cyclin B/Cdk1 or MAPK.[4] In *Xenopus*, Polo-like kinase kinase-1 (xPlkk1) was cloned as another putative upstream kinase[136] and PKA was also reported to activate Plk1.[124] Like xPlkk1, related Ste20-like kinases were found to phosphorylate Plk1, i.e., human SLK can phosphorylate mouse Plk1[137] and mouse LOK can also phosphorylate Plk1 in vitro.[75] It remains to be determined whether Ste20-like kinases phosphorylate Plk1 in vivo and whether Cdk1, PKA and Mps1 can also regulate Plk1.

Possible target residues in Plk1 for upstream kinases are the highly conserved Ser-137 and Thr-210, of which the latter is located in the so-called activation or T loop. In both *Xenopus* and human cells, phosphorylation of Thr-210 was shown to be a major activating phosphorylation event.[117,124] When Thr-210, Ser-137 or both residues are mutated into Asp to mimic phosphorylation, Plk1 kinase activity is increased severalfold.[95,117] So far, no upstream kinase responsible for phosphorylating Ser-137 has been identified and conflicting data exist about the kinase upstream of Thr-210. In *Xenopus* and HeLa cells, xPlkk1 was found to phosphorylate Thr-210, but not Ser-137.[75,117] However, in another *Xenopus* study PKA but not xPlkk1 was shown to phosphorylate Thr-201, the Plx1 residue analogous to Thr-210.[124] No evidence could be obtained for phosphorylation of Ser-128 in Plx1, the residue corresponding to Plk1 Ser-137. All above-mentioned phosphorylation studies were performed in prometaphase-arrested cells, suggesting Ser-137 may be phosphorylated later in mitosis. With the use of phospho-specific antibodies, Thr-210 phosphorylation was indeed found to precede Ser-137 phosphorylation in vivo, the latter occurring specifically in late mitosis.[76] In contrast, another report shows that Ser-137 phosphorylation can also be detected in early mitosis, at the same time as Thr-210 phosphorylation.[138] However, previously performed peptide maps have not detected phosphorylation at Ser-137 early in mitosis.[75,124,136] Furthermore, functional experiments with mutants of this residue indicate a function for Ser-137 phosphorylation in the later stages of mitosis,[76] which suggests that Ser-137 phosphorylation only becomes important in late mitosis. Thus, Plk1 function may be modified by different posttranslational modifications that occur at distinct stages in mitosis.

Phosphorylation of Ser-137 and/or Thr-210 may also contribute to recovery from a DNA damage-induced G_2 arrest. Phosphorylation of these residues was reported to be inhibited in vivo after DNA damage.[138] Introducing a T210D single mutation or a S137D/T210D double mutation have previously been shown to render Plk1 insensitive to DNA damage-induced downregulation of activity.[18] Interestingly, expression of Plk1-T210D or Plk1-T210D/S137D can override the G_2 arrest induced by DNA damage,[18] indicating that Thr-210 and possibly also Ser-137 phosphorylation may be important for DNA damage checkpoint recovery. Alternatively, these data may reflect a general requirement for Plk1 activity in checkpoint recovery, since both mutants possess highly elevated kinase activity.

In recombinant Plx1, several other residues were identified as phosphorylation sites (Thr-10, Ser-25/26, Ser-326, Ser-340) and some of these residues were found to be phosphorylated by Plkk1 and Cyclin B/Cdk1 (Thr-10 and Ser-340, respectively).[124] Except for Ser-326, none of these residues are conserved in Plk1, thus leaving it unclear what residues in Plk1 are phosphorylated by upstream kinases. *Xenopus* Plx1 probably auto-phosphorylates itself on Ser-260 and Ser-326[124,139] and this may also be valid for Plk1 since both these residues are conserved. Plk1 can indeed auto-phosphorylate itself[2] and may thus enhance or inhibit its own activity. Binding of Plk1 to a phosphopeptide with an optimal binding motif increases Plk1 activity,[127] suggesting that Plk1 activity is also increased upon substrate binding. Furthermore, Plk1 activity can be amplified by binding to motifs phosphorylated by itself. Plk1 phosphorylates Thr-68 of Chk2[22] and Plk1 activity can be stimulated upon binding to a phosphopeptide encompassing phosphorylated Thr-68.[23] However, it seems unlikely that this takes place in vivo, since Plk1 is inhibited upon DNA damage while Chk2 is activated. Plk1 is also known to regulate its own recruitment to the midbody through creating its own binding site on MKlp2.[57] This process of self-amplified recruitment and activation is probably conserved with other substrates and/or at other subcellular sites and may comprise an important auto-amplification mechanism.

Plk1 recruitment to kinetochores was recently shown to depend on Cdk1-dependent phosphorylation of INCENP, a protein of the chromosomal passenger complex.[140] Mutation of the INCENP residue responsible for this event, Thr-388, results in a delay in the transition from metaphase to anaphase,[140] indicating the importance of proper Plk1 recruitment at kinetochores for mitotic progression. It was suggested that this involves direct binding of Plk1 to Thr-388-phosphorylated INCENP.[140] However, since INCENP is known to localize to centromeres whereas Plk1 is thought to reside at kinetochores, the exact localization of Plk1 at the centrosomere/kinetochore region needs to be further investigated. Also, weak Plk1 kinetochore localization can still be detected in some cells expressing the INCENP Thr-338 mutant, suggesting that other factors may also be involved in Plk1 targeting to this site.[140] Indeed, kinetochore localization of Plx1 was recently found to depend on xMps1,[46] indicating several upstream factors may cooperate in this process.

In response to DNA damage, Plk1 activity is inhibited, Plk1 protein is degraded and the Plk1 gene is repressed through the actions of several proteins. Plk1 activity is inhibited through the actions of ATM and/or ATR,[45] the Plk1 gene is repressed by BRCA1 and Rb family members p130, p107 and p105Rb,[21,141,142] and Plk1 levels are downregulated through Chfr-mediated ubiquitination of Plk1 and subsequent targeting to the proteasome.[143,144] The Plk1 promoter is cell cycle regulated through activation by Forkhead transcription factor FoxM1[145] and repressed by cell cycle-dependent element (CDE)/cell cycle-gene homology region (CHR) elements.[146,147]

In summary, Plk1 activity can be regulated through transcriptional regulation, ubiquitin-mediated degradation, (auto)phosphorylation and substrate binding. Furthermore, the general stability of Plk1 is regulated by Hsp90 and the chaperonin-containing TCP1 complex CCT.[148,149] Since the upstream regulation of Plk1 is the aspect of

Plk1 regulation that is the least understood, more research is needed to understand exactly how all the different functions of Plk1 are regulated by upstream processes.

Regulation of Plk2-4. Plk2 was originally identified as a mouse serum-inducible kinase and was hence named Snk.[81] Plk2 mRNA levels peak during G_1 and early S phase and are regulated by serum and the phorbol ester PMA.[81,83] Plk2 can probably auto-phosphorylate itself.[81] Furthermore, Plk2 is a p53 target gene and p53-dependent activation of Plk2 may prevent mitotic catastrophe after spindle damage.[86] In neurons, a Plk2/3-binding protein has been found, a calcium- and integrin-binding protein (Cib), which negatively regulates both Plk2 and Plk3 kinase activity.[82,83]

Mouse Plk3 was identified as a FGF-inducible kinase (Fnk).[91] Plk3 mRNA levels peak shortly after mitogen addition, but protein levels remain relatively constant during the cell cycle.[9,91-93] Plk3 mRNA expression is induced by FGF, platelet-derived growth factor, serum, interleukin-3, thrombopoietin and phorbol myristate acetate treatment of quiescent cells.[91,150] Since during most of the cell cycle Plk3 expression remains relatively constant, post-translational modification may be an important regulatory mechanism for Plk3. Mouse Plk3 is post-translationally modified upon serum stimulation and transiently phosphorylated in mitosis.[93] Like Plk1 and Plk2, Plk3 is capable of phosphorylating itself.[92] Plk3 is activated upon DNA damage in an ATM-dependent manner and this inhibition could involve Chk2.[9,20,104] Mitotic spindle disruption also upregulates Plk3 activity, but the underlying mechanism is unknown.[9]

Plk4 expression and kinase activity are dependent upon the cytoplasmic protein-tyrosine kinase Tec, which is activated by cytokine receptors, lymphocyte surface antigens, heterotrimeric G protein-linked receptors, and integrins.[134] Tec activity protects Plk4 from PEST sequence-dependent proteolysis.[134] Furthermore, Plk4 was identified as a p53 target gene and was shown to be repressed by p53.[108] It is likely that more upstream regulators of Plk2-4 will be found in the future, what will shed more light on the dark box of upstream regulation of these kinases.

Plks AND CANCER

Consistent with its role in cell division, Plk1 may contribute to the formation and/or progression of human cancers. Plk1 is overexpressed in many tumors such as melanomas, lymphomas and carcinomas in various tissues and was even reported to be a negative prognostic marker (for a review see Ref. 151). Overexpression of Plk in mouse fibroblasts results in loss of contact inhibition and such Plk-expressing cells can form tumors in nude mice.[6] These data indicate that Plk1 indeed plays a role in tumorigenesis.

If Plk1 can enhance tumor formation, then tumor regression may be achieved by selective inhibition of Plk1. The effect of Plk1 inhibition on cell proliferation and tumor growth has been assessed in several systems. Proliferation of several tumor cell lines is inhibited through apoptosis by a membrane-permeable part of the PBD, consisting of amino acids 410-429 of the PBD, containing the essential Trp-414, Val-415 and Leu-427 residues, fused to an Antennapedia peptide.[152] Similar results are obtained by Plk1 siRNA and anti-sense oligonucleotides.[153,154] In human leukemia cells, apoptosis can be induced by inhibiting Plk1 with β-hydroxyisovalerylshikonin (β-HIVS) isolated from a Chinese plant, but the specificity of this inhibitor was not investigated.[155] Apoptosis of tumor cell lines can also be achieved by inhibiting Plk1 function through adenoviral delivery of a dominant-negative Plk1.[156] Interestingly, normal human mammary epithelial cells do not undergo mitotic catastrophe after similar treatment, indicating tumor cell-specific apoptosis.[156]

In vivo tumor regression in xenograft nude mouse models has been reported with siRNA against Plk1 and combined antisense therapy against Plk1 and Bcl-2.[157,158] Scytonemin, a non-specific ATP competitor isolated from microalgae, inhibits Plk1 and cellular proliferation and induces apoptosis.[159] However, scytonemin also inhibits Cyclin B/Cdk1 and Myt1 at the same concentration, suggesting that the observed effects are due to the inhibition of multiple kinases.[159] A small molecule inhibitor of Plk1 activity, ON01910, was recently identified, which competes for the substrate binding site of Plk1.[160] In vivo, ON01910 potently inhibits tumor growth in xenograft nude mouse models and can even induce complete tumor regression when combined with several chemotherapeutic agents. Normal human cell lines were found to be very resistant to the apoptotic effects of ON01910,[160] but the effects on normal cells were poorly studied compared to the effects on tumor cells. Since Plk1 inhibitors may induce apoptosis in a tumor-specific manner, ON01910 has entered Phase I clinical trials for cancer therapy.

Mutations in Plk1 have not been reported in human tumors, although one study has investigated the presence of Plk1 mutations in human tumor cell lines.[161] No mutations were found in the cell lines that were studied with high levels of Plk1. In contrast, in several solid tumor lines mutations in the PBD of Plk1 were found which disrupt binding to Hsp90, leading to destabilization of Plk1.[161] The mutations were not in any of the previously identified residues that are important for PBD function. Since downregulation of Plk1 is an exception among tumors, it will be interesting to examine whether Plk1 mutations are present in other tumor cells which overexpress Plk1.

No information is available on Plk2 and Plk4 expression in tumor cells. Plk3 levels are downregulated in the majority of the investigated tumor samples, although Plk3 upregulation has also been detected (reviewed in ref. 151). In contrast to Plk1, Plk3 may be a growth suppressor gene, since overexpression of Plk3 causes apoptosis in human tumor cell lines.[94,162] Since Plk3 primarily functions in stress-responsive pathways, reducing Plk3 may overcome a stress-induced proliferation arrest, thereby possibly contributing to genomic instability. Recently, reducing levels of Plk4 has been implicated in tumor formation. $Plk4^{+/-}$ mice showed a dramatically increased incidence of spontaneous liver and lung cancers.[110] Thus, in contrast to Plk1, Plk4 may be a tumor suppressor protein. Taken together, of the human Polo kinases Plk1 forms the most attractive target for anti-tumor therapy. Selectively inhibiting specific Plk1 interactions and/or functions may improve the efficiency of such therapies and reduce the potentially hazardous side effects. It would be interesting to specifically interfere with the Plk1 functions in recovery after a DNA damage-induced arrest, since the cellular response to DNA damage was found to be activated in a large variety of human tumors, but not normal proliferating cells.[163,164] In these tumor cells, replication errors were detected leading to double strand breaks. Cancer cells probably overcome a DNA damage-induced cell cycle arrest by inactivating checkpoint proteins, leading to proliferation and genomic instability. Since Plx1 in *Xenopus* is necessary for the termination of a DNA replication checkpoint response,[78] Plk1 may be important for this process in human cells as well. Specific interference with the function of Plk1 in DNA damage recovery may therefore prevent cancer cells to restart the cell cycle and become malignant, but leave normal proliferating cells unharmed. The outcome of clinical trials with Plk1 inhibitors and the identification of more selective inhibitors are therefore eagerly awaited.

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